



**TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC 2 2018**

CANDIDATE
NAME

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CENTRE
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INDEX
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H2 BIOLOGY

Paper 4 Practical

9744/04

**Monday 27 August 2018
2 hours 30 minutes**

Candidates answer on the Question Paper
Additional Materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift	
Laboratory	

For Examiner's Use	
1	/ 23
2	/ 13
3	/ 19
Total	/55

This document consists of **19** printed pages and **1** blank page.

Answer **all** questions

- 1 Vitamin C, or ascorbic acid, is a water soluble antioxidant that plays a vital role in protecting the body from infection and disease.

It is not synthesised by the human body and must be acquired from dietary sources such as fruits and vegetables because many plant cells contain water soluble ascorbic acid.

You are provided with an extract from plant cells, **P**, which contains ascorbic acid.

Visking tubing, **V**, is selectively permeable, similar to a cell surface membrane, so that some biological molecules will diffuse through the wall of the tubing.

You are required to investigate the diffusion of ascorbic acid from **P** into the water surrounding the Visking tubing over a period of 15 minutes.

Fig. 1.1 shows the apparatus before the water was added.

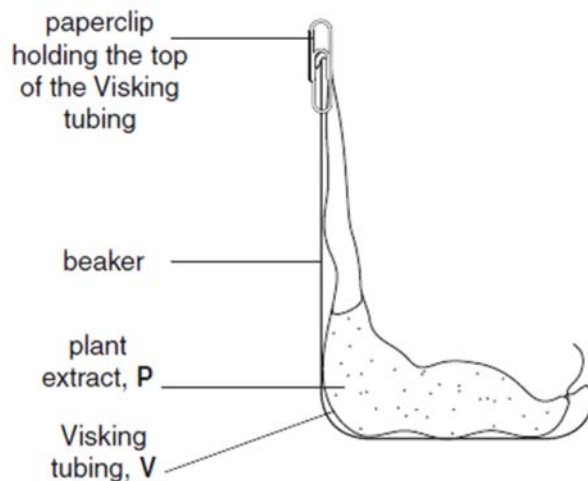


Fig. 1.1

- (a) (i) Water is added to the beaker in Fig. 1.1.

Describe the expected trend in the concentration of ascorbic acid in the water over a period of 15 minutes. [1]

- As time increases, concentration of ascorbic acid in the water increases.

You are provided with:

Labelled	Contents	Hazard	Volume / cm ³
A	sample of water removed after 15 minutes	irritant	15
P	plant extract containing ascorbic acid	irritant	15
W	distilled water	none	100
I	iodine in potassium iodide solution	irritant	20
S	starch	none	20

Labelled	Details
V	15 cm length of Visking tubing in a beaker containing water

You must now read up to the end of step 23 before proceeding.

To compare the concentration of ascorbic acid in the samples you are required to find the **volume** of iodine solution, **I**, added to each sample until the end-point is reached.

The drops of **I** will be added one at a time using a small syringe.

To practise releasing drops from a small syringe:

1. Fill the syringe with 2 cm³ of **I**.
2. Hold the syringe over an **empty** test-tube and push the plunger **gently** to release one drop at a time, as shown in Fig. 1.2.

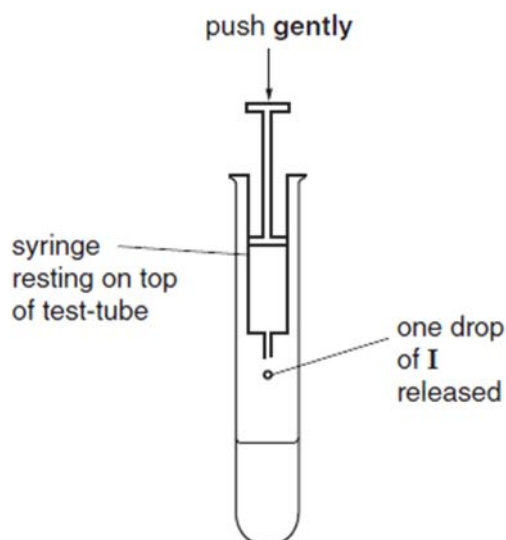


Fig. 1.2

To compare the concentration of ascorbic acid in the samples you will need to add drops of **I** until a blue colour appears. When this blue colour lasts for more than 10 seconds, this is the end-point and the **volume of I** that has been added should be recorded.

The apparatus in Fig. 1.1 was set up and water was added and left for 15 minutes.

Sample **A** was removed from the water in the beaker.

You are required to find the volume of **I** needed to reach the end-point for sample **A**.

Proceed as follows:

3. Put 1 cm³ of **S** into a test-tube.
4. Put 3 cm³ of the sample (e.g. **A**) into the same test-tube.
5. Shake the test-tube gently to mix the contents.
6. Fill the syringe, labelled **I**, with 2 cm³ of **I**.
7. Wipe off any drops of **I** from the outside of the syringe with a paper towel.
8. Add **one** drop of **I** to the mixture in the test-tube as shown in Fig. 1.2.
9. Mix gently and if there is no colour change add another drop.
10. Continue adding drops, one at a time, until the blue colour appears.
Wait 10 seconds to see if the end-point has been reached. If the blue colour disappears then add another drop.
11. Repeat step 10 until the mixture stays blue for at least 10 seconds.

(ii) Record the **volume of I** needed to reach the end-point.

- Correct DP (1dp)
- Any reasonable volume

volume _____ [1]

You are required to:

- set up Visking tubing containing **P** as in Fig. 1.3
- decide the level of water to put into the beaker
- remove samples of the water surrounding the Visking tubing at **5 minute intervals** for 15 minutes
- compare the ascorbic acid concentrations in the samples.

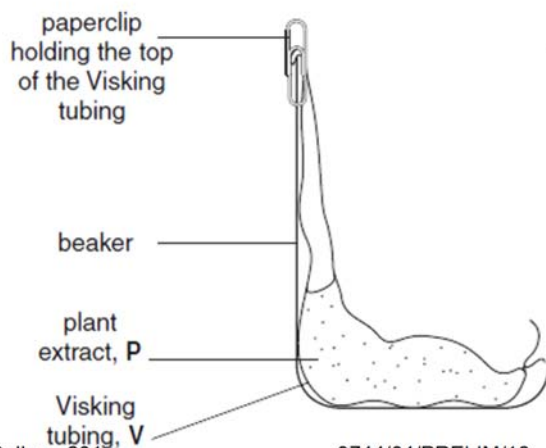


Fig. 1.3

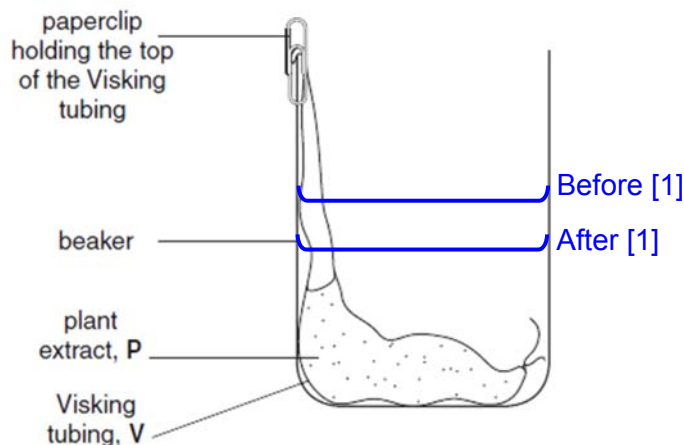
Samples of water surrounding the Visking tubing will be removed for testing, so you need to take this into account when you decide the level of water to put into the beaker.

(iii) Draw on **Fig. 1.3** the level of the water

- before you remove any samples (label 'before'),
- after the total volume of water needed for all the tests has been removed (label 'after').

[2]

- Two levels drawn labelled – one labelled 'before' & one labelled "after";
- The level of "after" lower than before;
- The level of "after" must still cover contents of Visking tubing ;



(iv) In order to compare the ascorbic acid concentrations, state **one** variable which you will need to standardise when finding the volume of **I** added to each sample.

Describe how you will standardise this variable.

variable :

- **volume of sample**
- OR
- **volume of starch**

description [1]

- **3cm³ of sample measured with a syringe**
- OR
- **1cm³ of starch added to the same test tube; measured with a syringe**

Proceed as follows:

- Put **S**, as in step 3, into the four test-tubes you will require in order to test the samples of water.
- Tie a knot in the Visking tubing as close as possible to one end so that it seals the end.
- To open the other end, wet the Visking tubing and rub the tubing gently between your fingers.
- Put 6 cm³ of **P** into the open end of the Visking tubing.

16. Rinse the outside of the Visking tubing by dipping it into the water in the container labelled **V**.
17. Put the Visking tubing into an empty beaker as shown in Fig. 1.3.
18. Make sure the open end of the Visking tubing is held in place by a paperclip.
You will start timing as soon as you add **W** (**steps 19 and 20**).
You should read steps 19 to 23 before proceeding.
19. Put **W** into the beaker to the level you decided in (iii).
20. Immediately start timing and remove the first sample of water (as in **step 4**) and put into a prepared test-tube (as in **step 12**).
21. Test the sample as in **steps 5 to 11**.
22. After 5 minutes, gently mix the water surrounding the Visking tubing and then remove the next sample, put it into a different (prepared) test-tube and repeat **steps 5 to 11**.
23. Repeat **step 22** for one more sample.

Record your results in (a)(v) on page 6.

(v) Prepare the space below and record your results. [5]

Time / min	volume of iodine to reach end-point / cm ³		
	Trial 1	Trial 2	average
0	0.2	0.2	0.2
5	0.8	0.9	0.9
10	1.4	1.4	1.4
15	1.9	2.0	2.0

CH: Column heading with UNITS – 1 mark

D : Different volume of iodine for different time interval + data for 4 different time intervals 1 mark

R : 2 readings – 1 mark

Tr : Trend – increase in vol of iodine as time increase – 1 mark

Pr : all values to one decimal place – 1 mark

(vi) Describe how the results support your expected trend as stated in (a)(i). [1]

- Answer in agreement with answer to (a)(i)

This investigation provides results to compare the concentration of ascorbic acid in the samples.

(vii) If you had been provided with 1.0% ascorbic acid solution, suggest how you would modify this investigation to find the **percentage concentration** of ascorbic acid in the water after 15 minutes. [4]

1. perform a serial dilution or simple dilution of 1% to obtain a total of 5 concentrations of ascorbic acid

2. find the volume of iodine to reach end point for each concentration
3. plot percentage concentration on X-axis and time taken on the Y-axis
4. time taken to reach end point for each sample of water is located on the graph and read off percentage concentration from the X-axis

(b) Iodine solution (iodine in potassium iodide solution) turns blue-black when starch is present in plant tissues.

However, as ascorbic acid is also found in plant tissues, some scientists investigated the effect of testing for starch with iodine solution when there was ascorbic acid present.

The concentration of ascorbic acid was $0.0001 \text{ mol dm}^{-3}$ and the concentration of starch solution was standardised.

The percentage of starch which reacted with the iodine solution was measured.

The results are shown in Table 1.1.

Table 1.1

Volume of iodine solution / cm^3	Percentage of starch which reacted with iodine solution
0.0	0.0
0.5	2.0
1.5	5.0
2.0	36.0
2.5	68.0

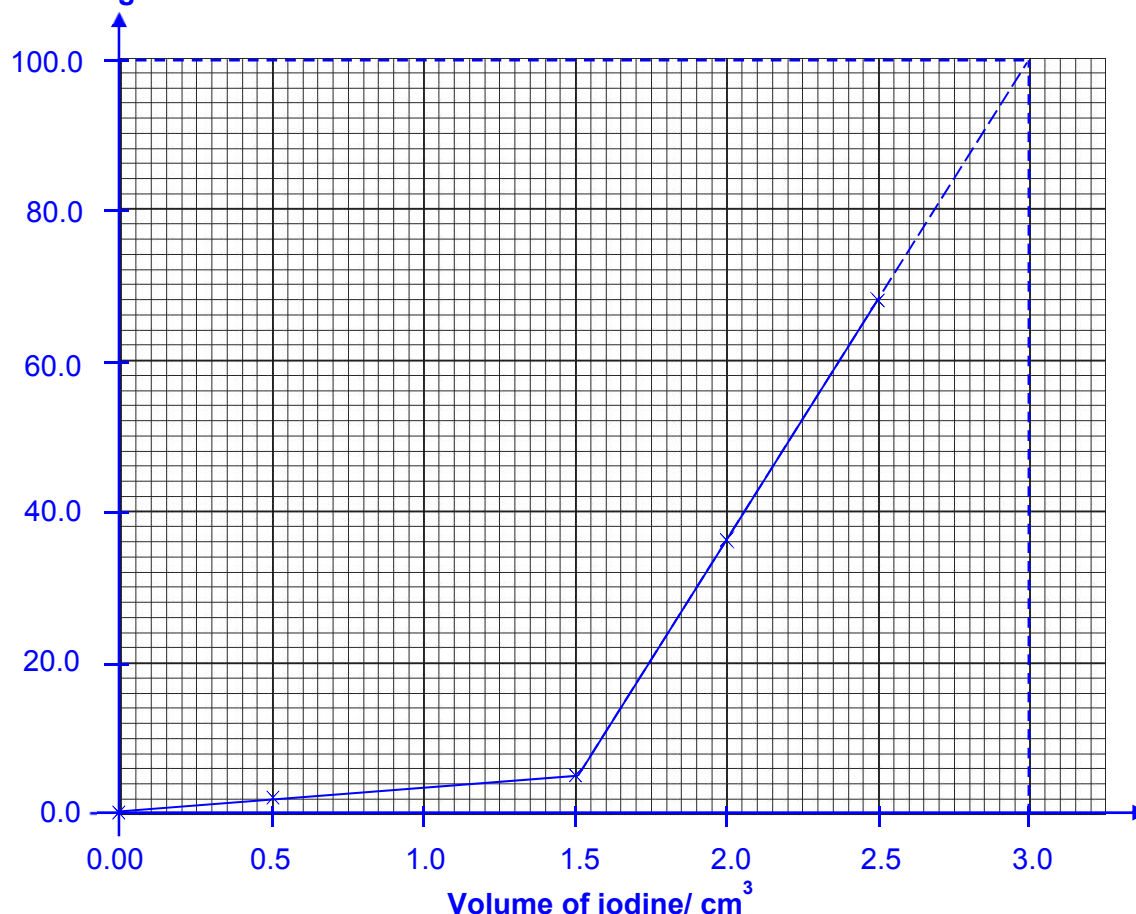
(i) Plot a graph of the data shown in Table 1.1.

You will need to **consider the answer to (b)(ii)** before you plot your graph. [4]

Plot a graph of the data shown in Table 1.1.

You will need to **consider the answer to (b)(ii)** before you plot your graph. [4]

Percentage of starch which reacted with iodine solution



NOTE: 1 mark EACH

S – Scale

- Mark axes at even interval.
- Drawn curve must cover 1/2 of graph paper (both x and y axes)
- Label origin (0,0) for BOTH axes

P – Plotted points:

- All points plotted accurately

A – Axes labelled :

- X axis: Volume of iodine solution/ cm³
- Y-axis: Percentage of starch which reacted with iodine solution

L – Line:

- All the points joined point-to-point

(ii) Estimate the volume of iodine solution needed for **100%** of the starch to be reacted.

Show on your graph how you obtained the volume of iodine solution. [1]

volume of iodine solution **3.0** cm³

Dotted lines on the graph to show how the value was obtained [½]

Volume stated [½]

- (iii) Explain how the presence of ascorbic acid may affect the use of iodine solution as a test for the presence of starch in different plant tissues. [2]

1. If there is too much ascorbic acid, all the iodine reacted with ascorbic acid and none left to react with starch. [1]
2. Need to find out the volume of ascorbic acid present before testing for starch to ensure that there is sufficient iodine available to react with starch. [1]

- (iv) A plant tissue contains $0.0001 \text{ mol dm}^{-3}$ ascorbic acid and starch.

Suggest how you would make sure that the iodine test showed the presence of all the starch (100%). [1]

1. Use 3cm^3 or more than 3cm^3 of iodine

[Total: 23]

- 2 Yeast cells have transport proteins in their cell membranes for the uptake of nutrients from the surroundings. There are **separate transport proteins** for **glucose** and for **maltose**. When exposed to both glucose and maltose the transport protein for maltose is downregulated and is not produced.

Plan an investigation to find out whether or not the **yeast transport proteins** for **glucose** and **maltose** function at the **same rate**.

Glucose and maltose are **both reducing sugars**.

You are provided with the following materials. Choose your materials from this list.

You **may not use any additional materials**.

- 10% yeast suspension
- 10 g dm⁻³ glucose solution
- 10 g dm⁻³ maltose solution
- Benedict's solution
- dilute hydrochloric acid
- dilute sodium hydroxide solution and sodium hydrogencarbonate solution for neutralising
- beakers and flasks of different sizes
- stopwatch or electronic timer
- colorimeter and tubes
- centrifuge and centrifuge tubes
- thermometer
- thermostatically-controlled water baths
- pipettes and pipette fillers
- burettes and burette stands
- filter funnels and filter paper
- syringes
- glass rods for stirring
- test-tubes and boiling tubes
- test-tube and boiling tube racks

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- include a clear statement of the hypothesis or prediction
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 13 marks]

[Independent variable] IV [MUST HAVE+ repeat with glucose/ maltose in methods: 1]
Independent variable: Types of sugar (glucose, maltose)

[Dependent variable]

Dependent variable: Absorbance value of yeast-sugar mixture after conducting Benedict's test for reducing sugar

[Hypothesis/ Theory/ Trend]

Tr [1]

1. Glucose is a monosaccharide, whereas maltose is a disaccharide, which is longer / larger (made of 2 α -glucose monomers) so it takes longer to transport maltose into the cell.
2. Furthermore, glucose is the preferred respiratory substrate as compared to maltose.
3. Hence, the rate of uptake of glucose is faster than rate of uptake of maltose.
4. Less glucose is found in the surrounding solution, hence the colour intensity of the Benedict's test and absorbance value will be lower, as compared to maltose.

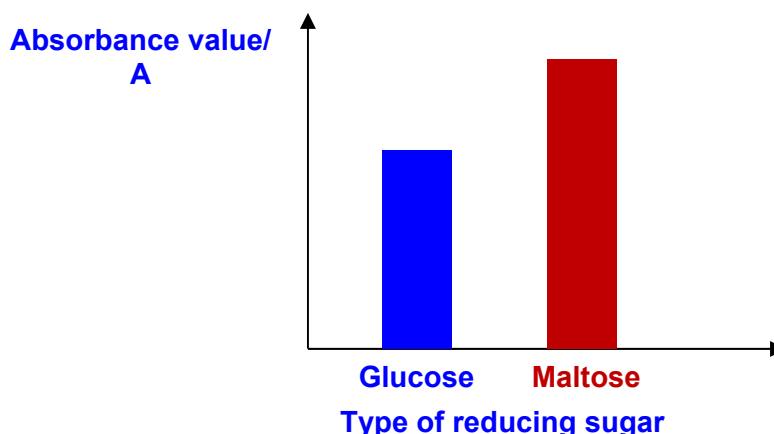
[Table]

T [1]

Type of sugar	Absorbance value/ A			
	Trial 1	Trial 2	Trial 3	Average
Glucose				
Maltose				

[Graph]

G [1]



METHOD:

[Constant variable: Volume of yeast, sugar]

V [1]

1. Using a 5mL syringe, add 2.0cm³ of yeast into a boiling tube.
2. Using a 5mL syringe, add 2.0cm³ of 10.0 gdm⁻³ glucose into another boiling tube.

[Equilibrate]

Eq [1]

3. To equilibrate, incubate (both) boiling tubes separately in a thermostatically controlled water bath at 30°C for at least five minutes.

[Constant variable: Temperature]

T° [1]

4. Mix/ Transfer the glucose to the boiling tube of yeast.
5. Immediately place the boiling tube back into the thermostatically controlled water bath at 30°C and start the stopwatch.

[Constant variable: Time]

Ti [1]

[Filter]

F [1]

6. After 5 minutes, remove the boiling tube and filter the content of the boiling tube using filter paper and filter funnel.

[Alternative: Pour the content of the boiling tube into the centrifuge tube and place it in a centrifuge]

[Benedict's test]

BT [MUST HAVE: 1]

7. Obtain/ Extract 2.0 cm³ of filtrate and add it to 2.0 cm³ of Benedict's solution in a boiling tube.

[Accept: Supernatant/ fluid component obtained in step 6]

8. Place the boiling tube in a boiling water bath for 2 minutes.

[Data collection]

DC [MUST HAVE: 1]

9. Transfer 2.0 cm³ of the mixture to a colourimeter tube (i.e. cuvette) and place it into a colourimeter.

10. Record the absorbance value of the mixture.

[Accept:

- Time taken for mixture to change colour from blue to green
- Mass of reducing sugars]

[Repeats for maltose]

IV

11. Repeat steps 1 to 10 for maltose solution.

[Repeats/ Triplicate]

RT [1]

12. Repeat steps 1 to 11 to obtain a total of three readings (triplicate) for using fresh samples of yeast, glucose, maltose.

[Rate]

R [MUST HAVE: 1]

13. The rate of uptake of sugars can be obtained from the gradient of the graph.

[Accept: $1 \div \text{time taken}$, where appropriate]

[Control]

C [1]

Control: Replace glucose or maltose with distilled water, and repeat the experiment, subject to the same experimental conditions.

[Safety Precaution]

S [1]

PRECAUTION	RISK
1. Do <u>not</u> handle <u>colourimeter</u> with <u>wet hands</u>	to <u>prevent electrocution</u> (<u>high risk</u>).
2. Use a <u>mitten/ insulated gloves</u>	to <u>prevent scalding/ burns</u> when using the <u>boiling water bath</u> (<u>medium risk</u>).
3. <u>Wear safety goggles</u> and <u>gloves</u> to avoid contact with <u>eyes</u> and <u>skin</u>	as <u>yeast</u> is a microorganism/ biohazard (<u>medium risk</u>). OR as <u>glucose/ maltose/ Benedict's solution</u> is an <u>irritant</u> (<u>medium risk</u>).

- 3 The eyepiece graticule scale in your microscope may be used to measure the actual length of the layers of tissues or cells, if the scale has been calibrated against a stage micrometer.

- (a) Using the stage micrometer, where one division is **0.1 mm**, calculate the actual length of one eyepiece graticule unit of 10X objective by completing step 1 and step 2.

Step 1

$$1 \text{ eyepiece graticule unit} = 10 \times 0.1 \text{ divided by } 100 = 0.01 \text{ mm [1]}$$

Step 2

Convert the answer to a measurement with the unit most suitable for use in light microscopy.

$$0.01 \text{ multiplied by } 10^3 \text{ [1]} = 10 \mu\text{m [1]}$$

[3]

(b) Slide **T1** is a transverse section of a leaf.

You are not expected to be familiar with this specimen.

- (i) Select a field of view so that you can observe and draw a large plan diagram of the part of the leaf indicated in Fig. 3.1.

You are required to use a sharp pencil for drawings.

Use **one** ruled label line **each** to label one layer of epidermis, xylem tissue and phloem tissue.

You should include only two vascular bundles.

You are expected to draw the correct shape and proportions of the different tissues.

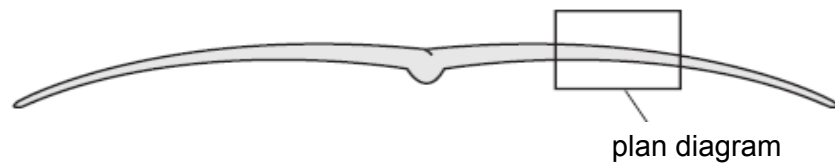
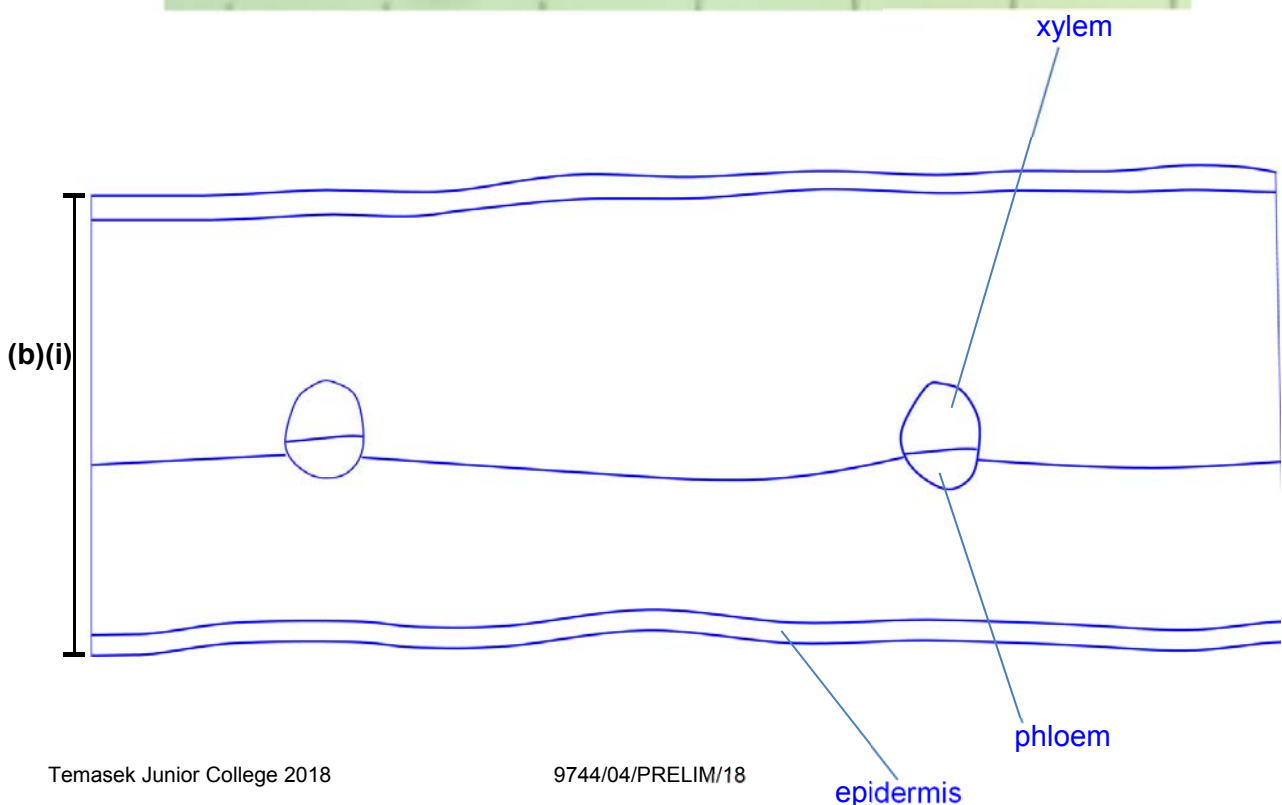
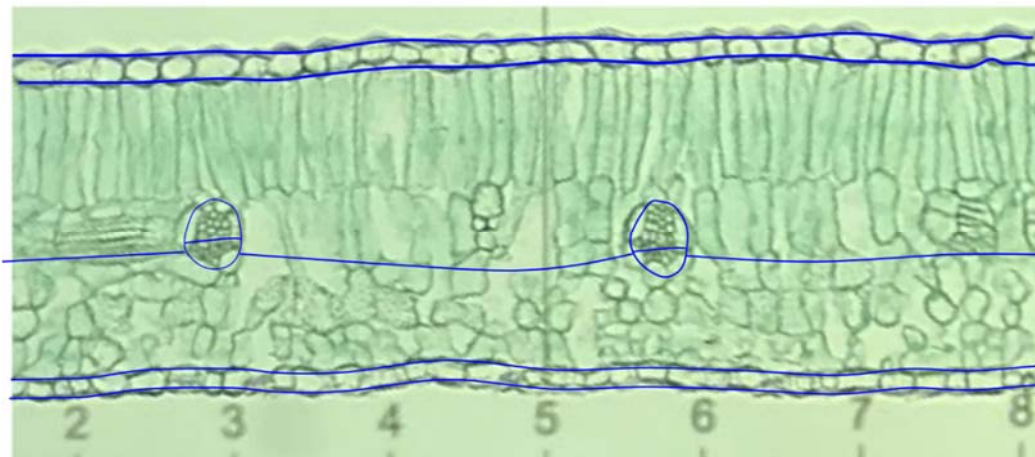


Fig. 3.1



M 1	1. clear, sharp, unbroken lines AND 2. no shading AND 3. LARGE SIZE [FILL MOST OF THE SPACE] AND 4. Must not use ruler for any part of the drawing (except labelling lines)	Reject - if drawn over the print of question - feathery lines - overlaps or gaps - any lines thicker than 1mm
M 2	1. no cells drawn AND 2. only correct section drawn AND 3. the 2 vertical lines at the left and right boundary must be drawn with ruler	Note: ▪ Must not draw box ▪ Vertical lines must not exceed the upper and lower epidermis
M 3	1. upper epidermis and lower epidermis drawn with two lines with distance 3 mm or closer for most of length. 2. Use a line to separate palisade and mesophyll layer 3. 2 vascular bundles drawn [correct shape]	Reject - if epidermal layers are too thick - if circles are drawn representing xylem vessels - if palisade layer is thinner than mesophyll layer
M 4	1. correct label with label line to epidermis, xylem and phloem tissues	Reject - if any label is biologically incorrect e.g. regions belonging to other organs or animals. • label within drawn area

[4]

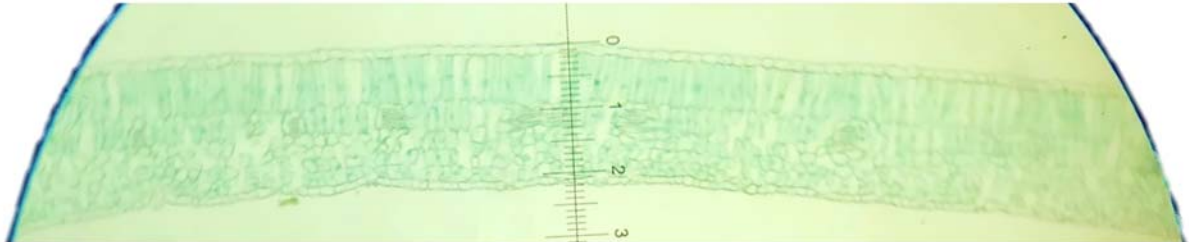
- (ii) Using the eyepiece graticule, measure the actual thickness of the leaf lamina, under low power.

Measure the size of your drawing across the same point.

Draw a line on your drawing in (b)(i) to show where you made the measurement.

Calculate the magnification of your drawing.

Show your working.



Measurement of leaf lamina = 22 div

Actual size of leaf lamina = 22 X 10 μm

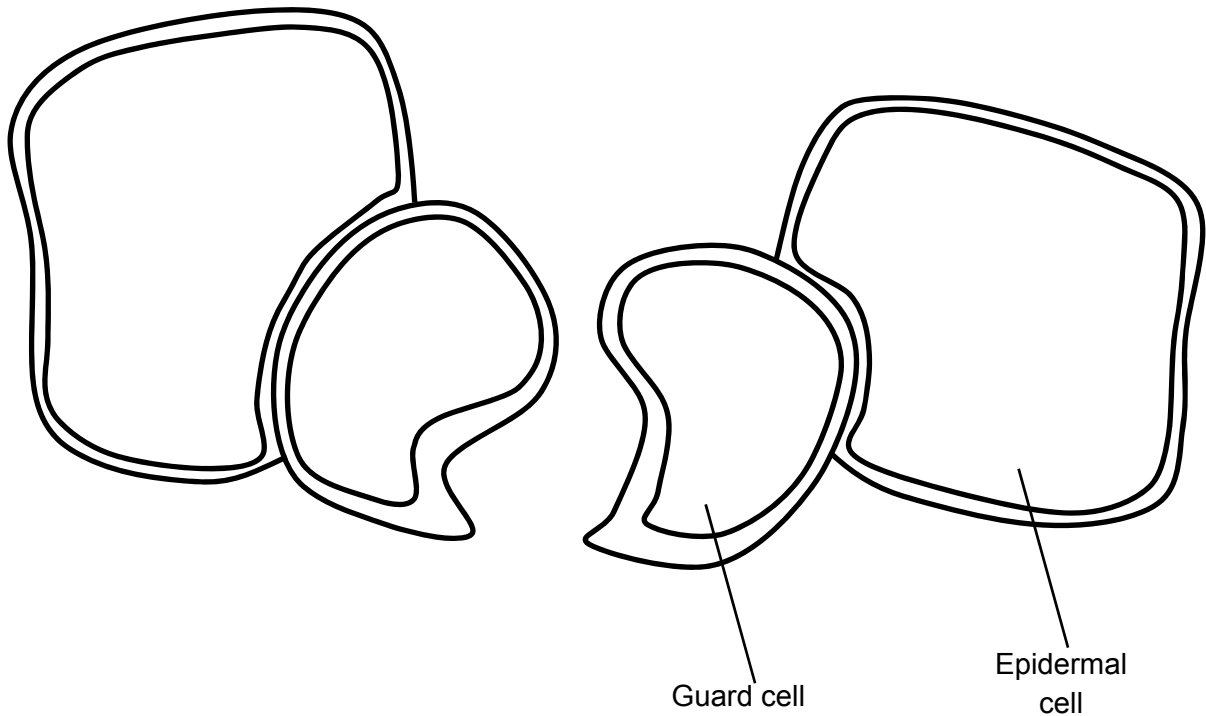
Length of drawing = 100mm

$$\begin{aligned} \text{Magnification} &= \frac{\text{Size of image}}{\text{Actual size}} \\ &= \frac{100 \times 10^3 \mu\text{m}}{22 \times 10 \mu\text{m}} \end{aligned}$$

magnification **X454** [2]

- (iii) Stomata are found on the lower surface of the leaf in the specimen on slide **T1**. The pores are likely to be closed and you may have to search to find a clear example.

In the space below, make a high-power, labelled drawing of **two** guard cells and their adjacent epidermal cells.



M 1	1. clear, sharp, unbroken lines AND 2. NO shading AND 3. Use up as much of the space as possible for drawing the cells	Reject - if drawn over the print of question - feathery lines - overlaps or gaps - any lines more than 1mm
M 2	1. Cell walls drawn as double lines. 2. Separated by a space not more than 2mm.	Cells must be large
M 3	1. 2 guard cells (correct shape) AND 2. 2 epidermal cells 3. Guard cell must in contact with 1 epidermal cell	Reject: Guard cells drawn as 2 bean-shaped cells
M 4	1. Label guard cells & epidermal cells 2. Use ruler to draw label lines	

[4]

- (c) Fig. 3.2 is a photomicrograph of a stained transverse section through another species of leaf.

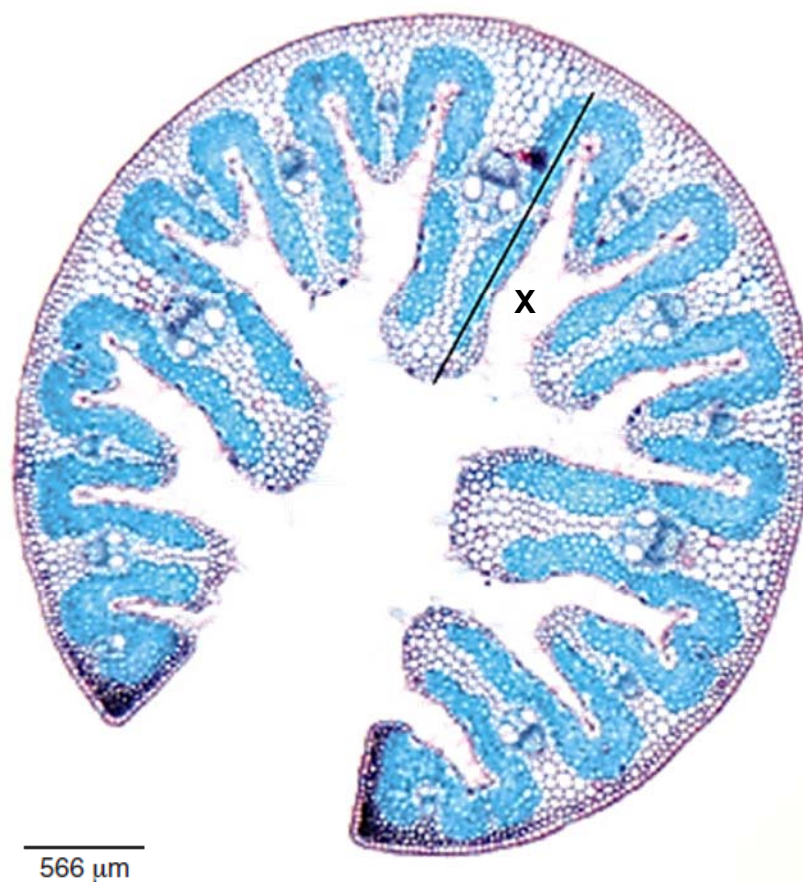


Fig. 3.2

- (i) Calculate the actual length of the fold shown by line **X**, using the scale bar.

You may lose marks if you do not show your working or if you do not use appropriate units.

Scale bar = 16mm

16mm $\cong$ 566 μ m [1]

Length of line = 44mm = 44 X 10³ μ m

Actual length = $\frac{44 \times 10^3 \times 566 \mu\text{m}}{16 \times 10^3 \mu\text{m}}$ [1] = 1556.5 μ m

actual length **1556.5 μ m [1]**

Fig. 3.3 is a magnified section of the leaf in Fig. 3.2.



Fig. 3.3

- (ii) Prepare the space below so that it is suitable for you to compare and contrast the location of stomata observed in specimen **T1** with Fig. 3.3.

Record your observations in the space that you have prepared.

Feature	T1	Fig. 3.3
Occurrence of stomata	Stomata only located on one epidermis for both	
Location of stomata	Lower epidermis	Upper epidermis
Location of stomata	Stomata exposed to external environment	Stomata hidden in folds of leaf
Frequency of stomata	Stomata evenly spread out	Not spread out, only occurs at folds
Presence of trichomes	Stomata not located next to trichomes	Located next to trichomes

[3]

[Total: 19]